CIFAR-10 Classification Challenge - Report

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Introduction

This project focuses on building and evaluating machine ning models for image classification using the CIFAR-10 dataset, a standard benchmark in computer vision. Compared to the FashionMNIST dataset used in earlier course tutorials, CIFAR-10 contains natural RGB images with significant variation in background, lighting and object orientation, making the task more challenging.

The goals of this assignment are :

1. To construct a complete deep learning pipeline for image classification
2. To train both a baseline CNN and an improved CNN
3. To evaluate model performance rigorously using accuracy, loss curves and confusion matrices
4. To analyze the strengths and weaknesses of each model

CIFAR-10 Dataset Overview

CIFAR-10 consists of 60,000 colour images sized 32 x 32 x 3, evenly split into 10 classes : airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck

Dataset split used for this project :

- Training set : 50,000 images
- Validation set : 5,000 images (10% split from training)
- Test set : 10,000 images

The small size and low resolution of CIFAR-10 makes classification difficult, as fine-grained details are often hard to distinguish.

Data Exploration and Preparation

(I) Preprocessing

All images were transformed into PyTorch tensors and normalized using channel-wise statistics :

$$\mu=(0.4914, 0.4822, 0.4465), \sigma=(0.2470, 0.2435, 0.2616)$$

Normalization accelerates convergence and stabilizes training.

(II) Data Augmentation

To improve generalization, the following augmentations were applied to the training set :

- Random horizontal flip
- Random crop (padding = 4)

Augmentation forces the model to learn invariance to orientation and position, reducing overfitting.

(III) Dataloader Settings

- Batch size : 128
- Shuffle : True (train)
- num_workers = 0 (required for Windows multiprocessing stability)

This pipeline ensures efficient feeding to data to the GPU during training

Methodology

Two models were implemented :

1. A baseline CNN for reference
2. A deeper, regularized Better CNN inspired by VGG-style networks

(I) Baseline Model (SimpleCNN)

The baseline model contains two convolutional layers followed by two fully connected layers.

Architecture :

- Conv(3 → 32), ReLU → MaxPool
- Conv(32 → 64), ReLU → MaxPool
- Flatten
- Fully Connected : 4096 → 256
- Fully Connected : 256 → 10

This model has low capacity and minimal regularization, making it a suitable baseline for comparison.

(II) Final Model (BetterCNN)

The improved model incorporates deeper convolutional blocks, Batch Normalization and Dropout.

Architecture :

Block 1 :

- Conv(3 → 64) → BN → ReLU
- Conv(64 → 64) → BN → ReLU
- Maxpool

Block 2 :

- Conv(64 → 128) → BN → ReLU
- Conv(128 → 128) → BN → ReLU
- MaxPool

Block 3 :

- Conv(128 → 256) → BN → ReLU
- MaxPool

Classifier :

- Dropout(0.5)
- Linear(4096 → 512), ReLU
- Dropout(0.5)
- Linear(512 → 10)

Training Strategy

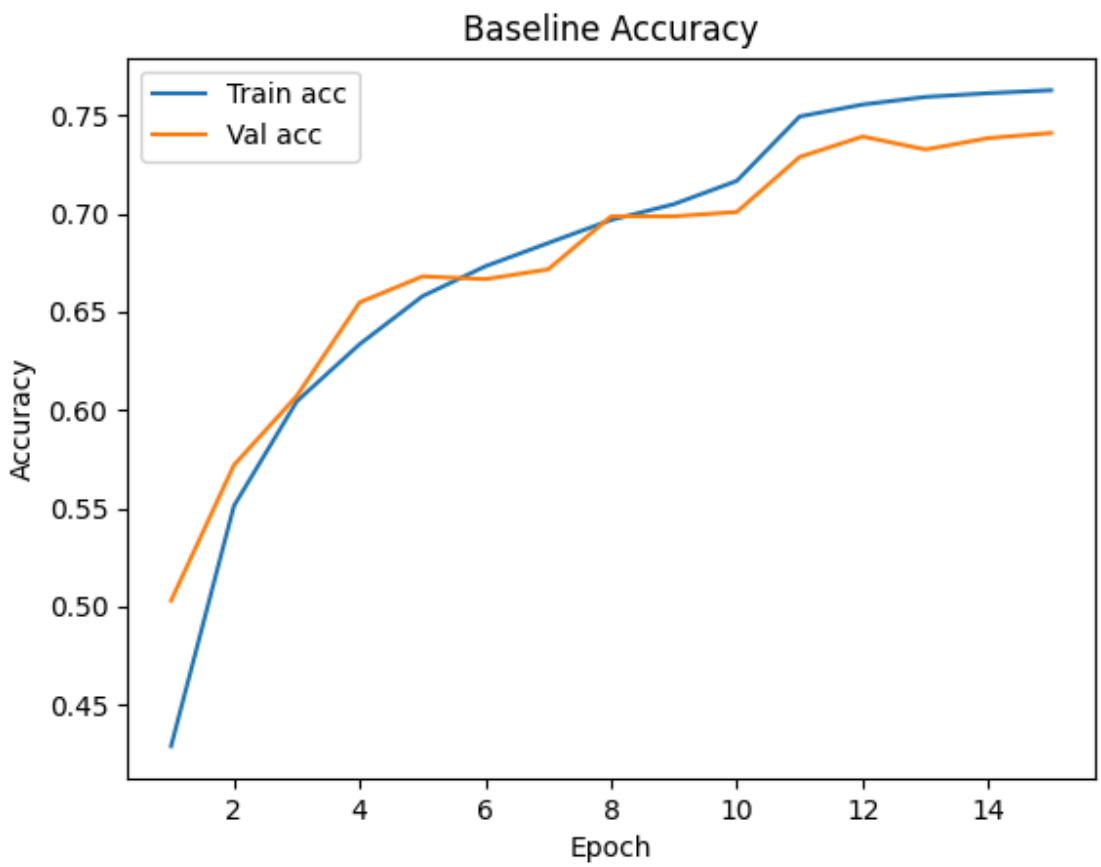
- Optimizer : Adam
- Learning rate : $1e^{-3}$
- Weight decay : 5 x 10^{-4}
- Learning rate scheduler : StepLR (step_size = 10, gamma = 0.1)
- Epochs :
    - Baseline : 15
    - Final model : 25

This combination balances expressive power with regularization.

Experiments and Results

(I) Baseline Model Performance

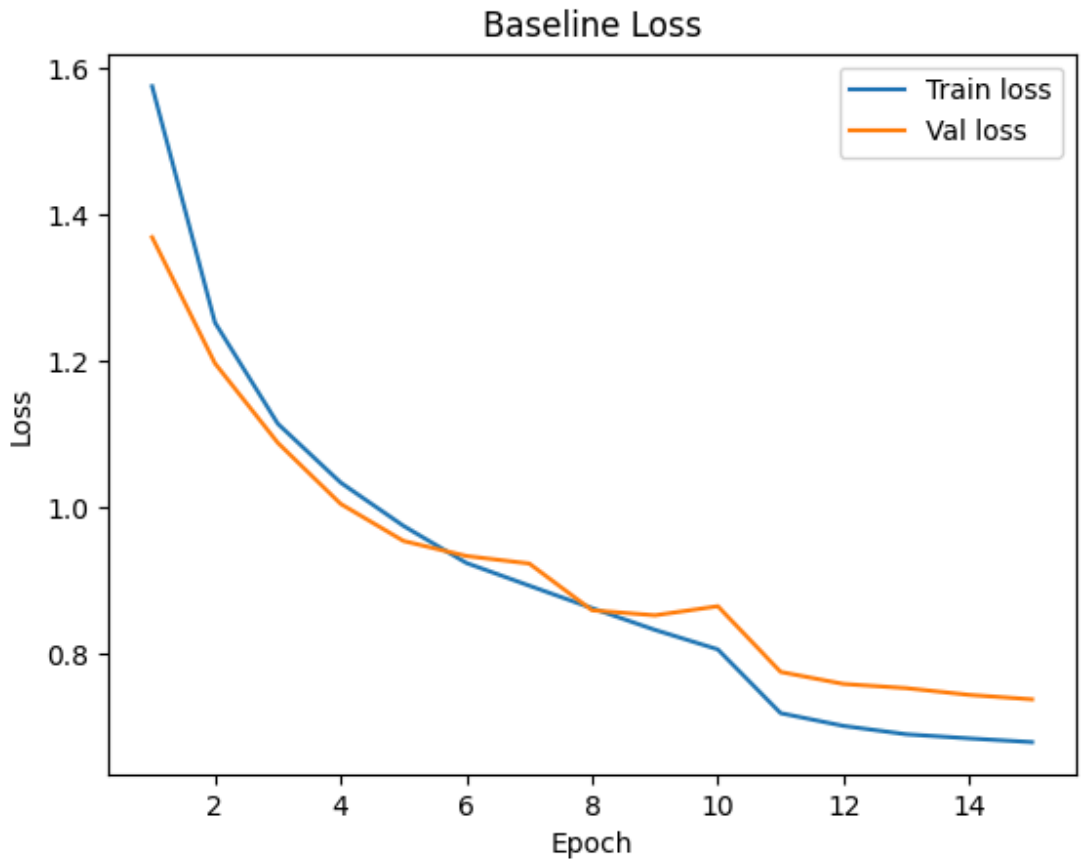
Baseline Accuracy Curve :



Observations :

- Accuracy rises steadily from ~42% to ~76%
- Validation accuracy closely matches training accuracy → low overfitting
- Performance plateaus around epoch 10 - 12

Baseline Loss Curve :



Observations :

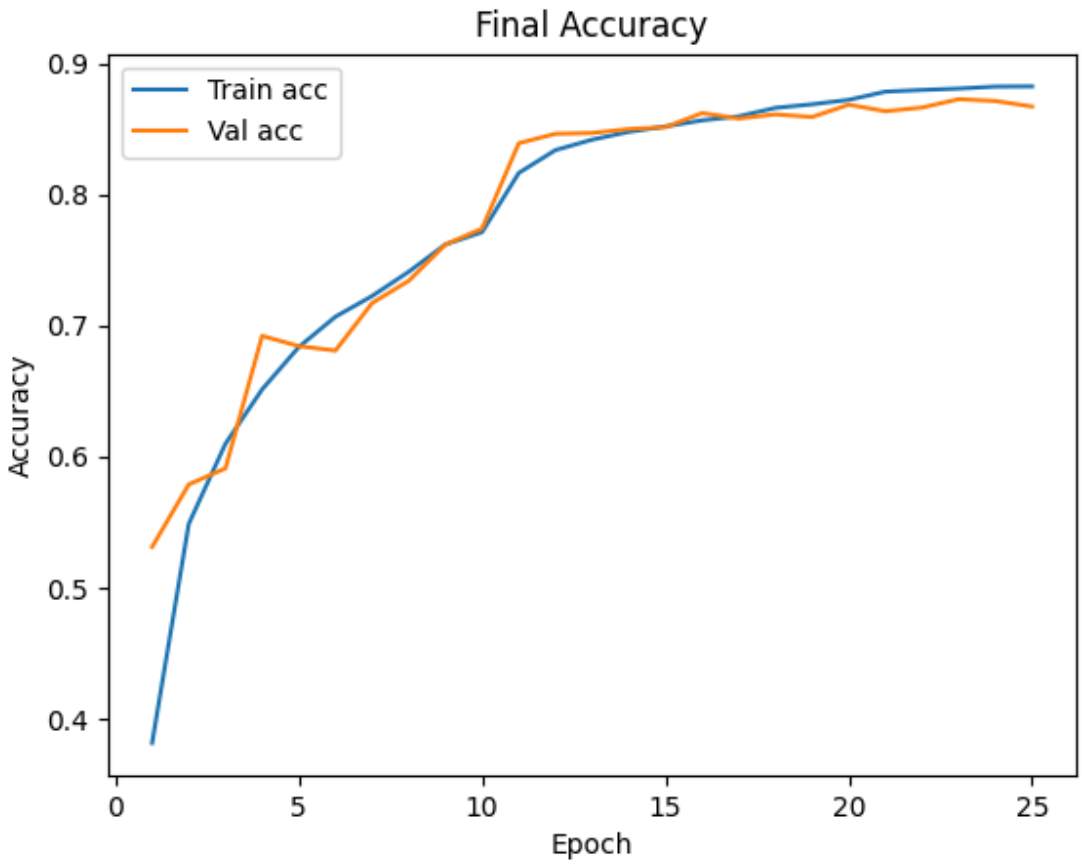
- Loss decreases smoothly, indicating stable training
- Validation loss begins to flatten around epoch 10

Baseline Test Accuracy :
76.33%

This represents typical performance for a shallow network without advanced regularization.

(II) Final Model Performance (BetterCNN)

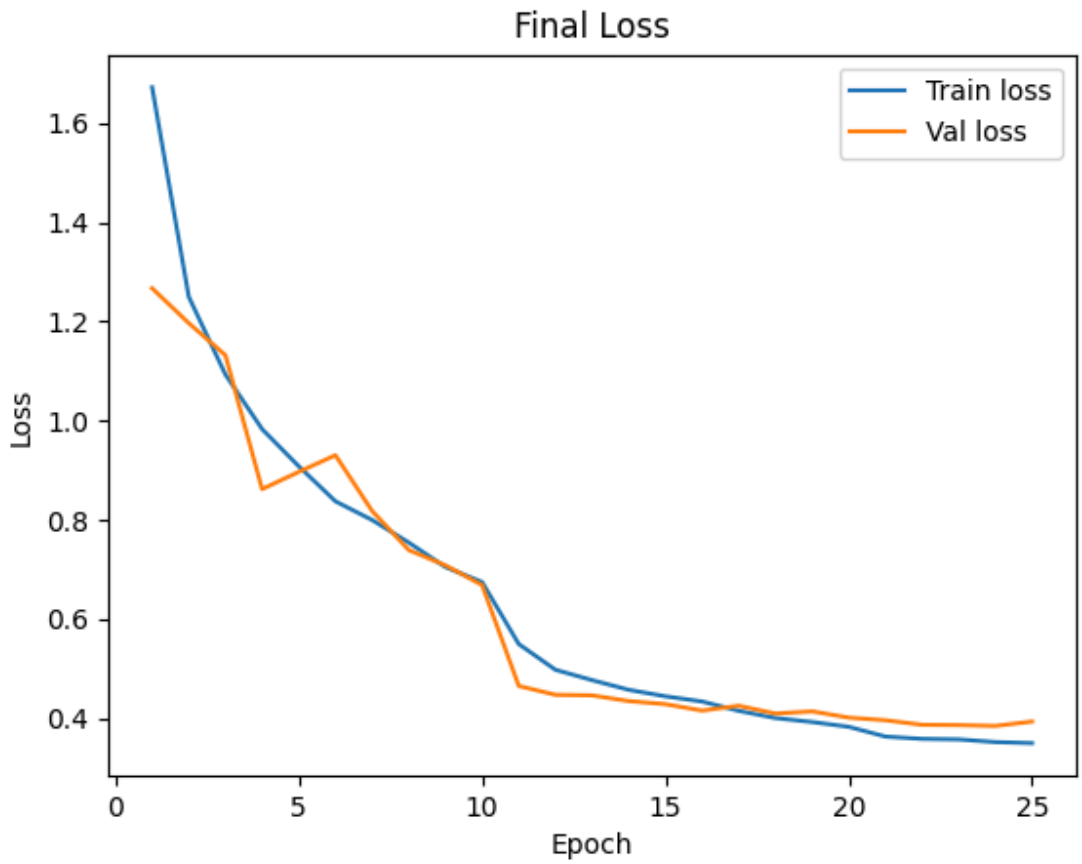
Final Accuracy Curve :



Key findings :

- Validation accuracy peaks at ~87%
- Training and validation curves remain close → excellent generalization
- Sharp improvement after epoch 10 due to LR scheduler lowering the learning rate

Final Loss Curve :



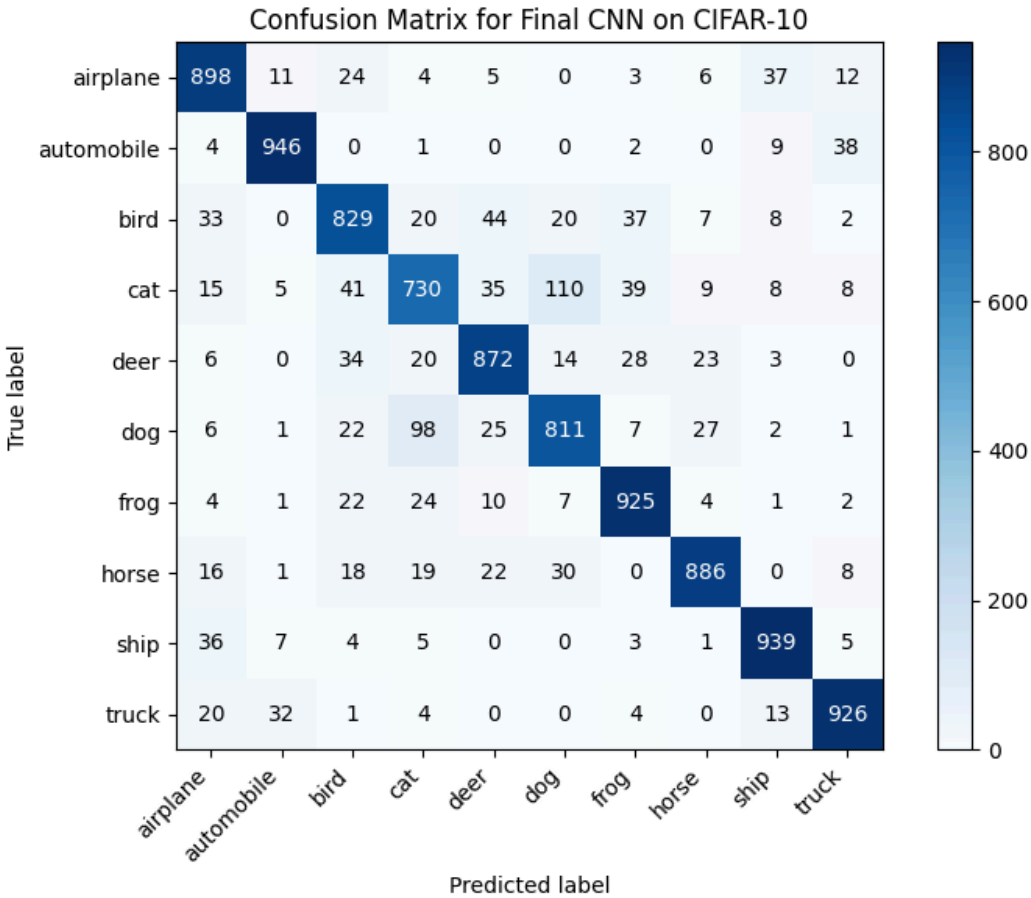
Observations :

- Substantial loss drop between epochs 10 - 12
- Final loss < 0.4 → strong convergence

Final Model Test Accuracy :
87.17%

This is a +10.8 percentage point improvement over the baseline model.

(III) Confusion Matrix Analysis



The confusion matrix reveals class-specific strengths and weaknesses.

Strongest Performing Class :

- Ship : 939 correct predictions
- Automobile : 946 correct predictions
- Frog : 925 correct predictions

These classes have distinctive shapes and colours.

Most Confused Class Pairs :

1. Cat ⇄ Dog
    - Cat → Dog : 110 misclassifications
    - Dog → Cat : 98 misclassifications
    - Reason : both are furry mammals with similar visual patterns at low resolution
2. Truck ⇄ Automobile
    - Truck → Automobile : 32 misclassifications
    - Automobile → Truck : 38 misclassifications
    - Reason : vehicle classes share similar silhouettes
3. Bird Confusions

    Bird is confused with :
    - Airplane (33)
    - Deer (44)
    - Frog (37)

    Small size and texture variation in bird images makes them hard to identify.

Overall, the model captures high-level structure well but struggles with fine-grained distinctions.

(IV) Qualitative Visualization

Observations :

- Most samples are classified correctly
- Correct predictions include difficult cases with cluttered backgrounds
- Misclassified images often appear visually ambiguous or noisy

This supports conclusions drawn from the confusion matrix.

Discussion

Why the Baseline CNN Underperformed :

- Limited Depth → shallow feature extraction
- No BatchNorm → unstable gradients
- No Dropout → risk of overfitting
- No LR rescheduling → slower convergence

Why BetterCNN Performed Strongly :

- Deeper architecture captures complex image patterns
- BatchNorm stabilizes training and accelerates learning
- Dropout prevents overfitting in dense layers
- Data augmentation increases robustness
- LR scheduling enables fine-tuning at later epochs

Overall Improvement :
76.33% → 87.17%
(+10.84% improvement)

This confirms the importance of architectural depth, normalization and training strategies in modern CNN design.

Conclusion

This project successfully implemented two CNN models for CIFAR-10 image classification and demonstrated the effects of architectural and training improvements on performance.

Key outcomes :

- Baseline model achieved **76.33%** test accuracy
- Final model achieved **87.17%** test accuracy, a significant improvement
- Confusion matrix revealed meaningful inter-class confusion patterns
- Training curves showed well-behaved optimization and low overfitting

This assignment provided practical experience in deep learning engineering, evaluation methodology and performance analysis - all essential skills in modern AI development.